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INSTRUCTIONS TO CANDIDATES

This **group assignment will contribute 50% to your final grade.**

Learning Outcome:

CLO3: Prepare a seminar portfolio that documents the student's activities that the students able to explain and analyse different analytics problems in the world of business

IMPORTANT

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Lecturer's Remark (Use additional sheet if required)

I Ooi Jin Yon (22039069), Liang Yrenn Da (22067128), Yeoh Jie Long (22071112), Vinnci Chin Wen Xi (22057269), Tan Jie Ming (22086003) received the assignment and read the comments.

Academic Honesty Acknowledgement

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Abstract

Business analytics has become an important technique to help financial institutions improve decision-making through the analysis of customer and marketing data. This study aims to analyse the UCI Bank Marketing dataset to identify customer and campaign characteristics, detect data anomalies and develop predictive models for term deposit subscription. The dataset had involved 45,211 customer records and 17 demographic, financial and marketing-related variables collected from direct marketing campaigns conducted by a Portuguese banking institution. Descriptive analytics and anomaly analysis were performed using Python to identify trends, outliers, missing values and unusual campaign patterns. Predictive analytics was conducted in Python by developing and comparing four machine learning classification models, which include Logistic Regression, Decision Tree, Random Forest and XGBoost. Data preprocessing techniques had included categorical encoding, outlier analysis, class balancing using SMOTE and feature selection were applied to improve model performance. The models were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC and confusion matrix analysis. The findings show that business analytics effectively identifies operational inefficiencies and customer behaviour patterns that help on better marketing decisions. Based on the evaluated models, Random Forest trained using the SMOTE-balanced dataset achieved the best overall predictive performance in term on improving customer targeting, reducing unnecessary marketing costs, and enhancing the effectiveness of future direct marketing campaigns.

1.0 Introduction

Business analytics has become an important technique for organisations **to** transform large volumes of data into actionable insights that support their decision-making. In the banking industry, customer data are collected from daily transactions, demographic records and marketing campaigns. By analysing these data, banks can have a better understand on their customer behaviour and identify different customer segments to improve the effectiveness of their business strategies. To increase their profit, banks are increasingly relying on business analytics to optimise marketing performance, improve customer satisfaction and allocate resources more efficiently.

One of the marketing approaches used in the banking industry is direct marketing. The company with contact their customers through telephone calls, emails or WhatsApp to promote banking products and financial services. Although direct marketing enables banks to communicate directly with potential customers, its effectiveness is often limited because not all customers are interested in the products being offered. Consequently, banks may spend more time and resources to contact customers that does not want to subscribe. This causes the increase in marketing costs and lower campaign success rates. Therefore, it is important to identify the customer who are more likely to respond positively to marketing campaigns.

Therefore, analytical techniques are used to help banks better understand customer behaviour and improve marketing effectiveness. Firstly, descriptive analytics will provide insights from historical customer data by identifying their trends, patterns and anomalies through statistical analysis and visualisation. After that, predictive analytics will be use to create machine learning algorithms that estimate the likelihood of customer subscription. This allows banks to target their potential customers more effectively while reducing unnecessary marketing costs.

Thus, this study uses the UCI Bank Marketing dataset to analyse customer and campaign information using descriptive analytics, anomaly analysis and predictive analytics. Python are used to perform data visualisation and analysis, while four machine learning classification models are developed and

compared to identify the most suitable model that can predict customer subscription more accurate and with less bias.

1.1 Objectives of the Study

The objectives of this study are:

1. To analyse customer demographic, financial and campaign data using descriptive analytics techniques.
2. To identify anomalies within the banking dataset.
3. To develop and compare four machine learning classification models for predicting customer subscription to term deposits.
4. To recommend the most appropriate predictive model for improving future customer targeting and marketing decision-making

1.2 Scope of the Study

This study focuses on the UCI Bank Marketing dataset that had 45,211 customer records and 17 variables from direct marketing campaigns of a Portuguese banking institution. The analysis includes data preprocessing, descriptive analytics, anomaly detection and predictive analytics in this project. Python are used to perform descriptive analysis and data visualization. Besides that, Python is used to develop four classification models, which include logistic regression, decision tree, random forest and gradient boosting. The predictive models are evaluated using multiple classification metrics such as accuracy, precision, recall, F1-score and ROC-AUC.

2.0 Literature Review

2.1 Business Analytics Applications

Business analytics has become a critical technique for organisations who wanted to transform large volumes of data into actionable insights that help to improve their operational efficiency, customer engagement and strategic decision-making. In the banking industry, business analytics will used statistical analysis, data mining, machine learning and predictive modelling to support various business functions. This includes the customer relationship management, targeted marketing, risk assessment and campaign optimisation. In short, business analytics enables organisations to make informed decisions based on evidence from the data itself.

Gul and Al-Faryan (2023) found that data analytics will significantly improve the company's productivity by improving the speed and accuracy of decisions. They also showed how banks were able to analyse customer behaviour and market trends to optimize resource allocation that led to the improvement of overall business performance. They concluded that data-driven decision-making will help financial institutions to respond more effectively to dynamic market conditions and shifting customer expectations.

Besides that, business analytics also plays an important role in the customer analytic and personalised marketing. Torrens et al. (2022) proposed a machine learning-based banking platform to analyses the customer transaction data to identify potential sales opportunities and make recommendations for them. They found that predictive analytics enables banks to understand customer preferences and group them based on the characteristics. Consequently, customer analytics improves the customer satisfaction while increasing marketing effectiveness through personalised communication strategies.

In addition, Nasir et al. (2024) had research on supervised machine learning algorithms that help to predict the customer responses to marketing campaigns by using the data. Their research found that predictive models can accurately identify the customer who will respond to promotional offers. This will allow banks to optimise their marketing resources and improve the success rates. By targeting the customer by group, they can reduce marketing costs while maximising customer retention.

Furthermore, business analytics are not only relied on analytical techniques but also rely on organisational data strategies. Grassi et al. (2022) had examined how financial institutions use customer data to create value under the open finance framework. Their findings suggest that integrating customer analytics with organisational data strategies enables banks to provide personalised financial services, strengthen customer relationships and improve competitive advantage. The study also highlights that data sharing and advanced analytics contribute to innovation in financial products and services.

Last but not least, research from Storey and Cohen (2002) supports the importance of predictive analytics for campaign optimisation in banking. The authors proposed an optimisation framework that uses existing response models and historical customer data to find the most profitable product offers for customers. Their findings show that predictive response models with optimisation techniques allows banks to allocate marketing resources more efficiently, which lead to improve campaign profitability and maximise marketing return on investment. This study established one of the foundational approaches for applying business analytics to marketing optimisation within financial services.

Overall, business analytics provides substantial benefits to multiple banking functions, which include customer analytics, targeted marketing, campaign optimisation and strategic decision-making. The use of predictive modelling, machine learning and data-driven decision-making will help organisations to have a better understand on their customer needs, enhance their marketing performance and improve the efficiency. In short, business analytics continues to serve as a key tool of competitive advantage and sustainable business growth.

2.2 Similar datasets/problems

Predictive analytics has become an increasingly important application of machine learning in the banking industry, enabling financial institutions to analyse large volumes of customer data and support data-driven decision-making. Banking organisations continuously collect demographic, financial and behavioural information to create structured data sets for predictive modelling. Although this project focuses on predicting customer subscription to term deposits using the UCI Bank Marketing dataset, previous studies have applied machine learning to a range of banking prediction problems, including customer response prediction, customer churn prediction, loan default prediction and credit risk assessment. Examining these studies provides insight into the characteristics of banking datasets and their suitability for predictive analytics.

Hou (2022) investigated customer response prediction by developing machine learning models to predict customers' likelihood of subscribing to bank term deposits using demographic, financial and marketing campaign information. Similar to the current study the data were structured customer information gathered in direct marketing campaigns. The data set is slightly different than the dataset from UCI Bank Marketing, but both studies have the goal of predicting customer subscription behaviour, showing that marketing data is still relevant for banking prediction tasks.

One of the common use cases of machine learning in the financial sector is predicting customer attrition. Lemos et al. (2022) utilised customer demographic, transactional and behavioural information from a Brazilian financial institution to predict customer churn. Likewise, Singh et al. (2023) have studied and analysed customer characteristics and customer behaviour at a retail banking environment and also introduced visual analytics to assist the customers in the interpretation of results. While the problem of customer churn prediction is not identical to the prediction problem for term deposits, it is a binary classification problem using structured customer data, and both are typical of banking applications that can be solved using supervised learning.

Although customer churn prediction is commonly studied in the banking industry, similar predictive analytics problems have also been investigated using datasets from other domains. Mouli et al. (2024) utilised a telecommunications customer churn dataset containing customer demographic, service usage and account-related information to predict customer churn. While the application domain is different from banking, both studies use structured customer data to address a binary classification task for customer behaviour. This shows that predictive analytics techniques are applicable to other industries and further support the fact that structured customer data is well suited for supervised learning applications.

In addition to the customer behaviour prediction, machine learning is also being used in banking risk management. Li et al. (2024) proposed a graph convolution-based model for loan default prediction using loan application data, while Chen (2022) applied machine learning techniques to analyse customer financial information for default loan prediction. While the studies concentrate on credit risk, and not on marketing results, they also make use of structured financial and customer data, showing that the fact that banks have vast amounts of data for different marketing tasks makes their datasets suitable for a wide range of predictive tasks.

Collectively, these studies show that banking prediction problems commonly utilise structured demographic, financial, transactional and behavioural information to support machine learning applications. Although the datasets are used to achieve different business goals, they have similar properties, and predictive models can be used to discover correlations among the attributes of customers and the business target.

The findings from previous studies therefore support the selection of the UCI Bank Marketing dataset for this project. It contains demographic, financial, and campaign and behavioural variables similar to those in previous banking prediction studies and thus is an appropriate dataset for testing machine learning models for term deposit subscription prediction.

2.3 Methods used

One of the most widely used methods in predictive analytics is machine learning, which allows models to learn from past information to predict future outcomes. In the studies reviewed, Logistic Regression, Decision Tree, Random Forest and XGBoost were the most common algorithms investigated, and preprocessing, feature engineering, class balancing, hyperparameter optimisation, and explainability were all recognised as factors in predictive performance.

Hou et al. (2022), Hon et al. (2023) and Chen (2024) compared multiple supervised machine learning algorithms for banking prediction tasks after applying appropriate data preprocessing techniques. Hou (2022) evaluated several machine learning models to predict customers' likelihood of subscribing to

term deposits using demographic, financial and marketing campaign data, illustrating the application of machine learning algorithms to banking marketing prediction. Similarly, Hon et al. (2023) compared multiple algorithms for banking customer churn prediction after applying preprocessing techniques such as removing irrelevant attributes, one-hot encoding and feature scaling. Although Support Vector Machine achieved the highest accuracy, XGBoost achieved the highest F1-score and ROC-AUC among the evaluated models. Similarly, Chen (2024) compared nine machine learning classifiers and determined the XGBoost model outperformed Logistic Regression, Decision Tree and Random Forest. Collectively, these studies demonstrate that comparing multiple machine learning algorithms enables researchers to identify the most suitable model for different banking prediction tasks while highlighting the importance of evaluating models using multiple performance metrics, including precision, recall, F1-score and ROC-AUC. Hon et al. (2023) further recommended the use of resampling techniques such as SMOTE in future work to improve predictive performance on imbalanced datasets.

In addition to algorithm comparison, Brito et al. (2024) and Peng et al. (2023) showed how the use of machine learning for effective preprocessing and optimisation techniques was crucial. Brito et al. (2024) have suggested a customer churn prediction model for retail banking which was based on data set with over 3 million customers and around 170 million transactions. The framework combines RFM-based feature engineering, ADASYN oversampling, NEARMISS undersampling, Bayesian hyperparameter optimisation and classification using XGBoost and Elastic Net. The implemented framework outperformed other customer churn prediction frameworks that were tested, demonstrating the importance of pre-processing for the customer churn prediction task. Similarly, Peng et al. (2023) combined Genetic Algorithm optimisation with XGBoost and SHAP analysis, reporting that the optimised GA-XGBoost model achieved the highest predictive performance among the evaluated models while SHAP provided explanations of the customer attributes contributing to churn predictions.

While not a banking dataset, the study by Mouli et al. (2024) still applies, as customer churn prediction in both the telecommunications and banking sectors are binary classification problems. The study compared Logistic Regression, Decision Tree, Random Forest and XGBoost, reporting Random Forest as the best-performing classifier. Other studies on banking-related topics, however, found XGBoost to be the best-performing model. The results show that the best predictive performance is not achieved by a single optimal algorithm, but is influenced by the characteristics of the datasets, the distribution of features and the applied preprocessing methods.

The following are common observations that emerge from the reviewed banking studies. The reviewed studies consistently incorporated preprocessing techniques such as feature encoding, feature scaling, feature engineering and class balancing before model development. In addition to overall accuracy, model evaluation was frequently carried out to determine precision, recall, F1-score and ROC-AUC, especially when dealing with imbalanced customer data. Recent studies have further incorporated optimisation and explainability techniques, including Genetic Algorithms, Bayesian hyperparameter optimisation and SHAP, to improve predictive performance and support interpretation of model predictions.

In summary, the literature reviewed indicates that the choice of machine learning algorithms is not the only factor affecting the performance of predictive models, but also the data preprocessing, feature engineering, optimisation, and evaluation techniques used. Guided by these findings, the present study employs Logistic Regression, Decision Tree, Random Forest and XGBoost to evaluate their predictive performance on the UCI Bank Marketing dataset for term deposit subscription prediction.

2.4 Visualisation & Analytics Methods

Business analytics applications in banking are built on solid descriptive analytics, anomaly detection and visual evidence for supporting strategic decisions. Before deployed advanced machine learning algorithms, organizations need to apply exploratory data techniques to understand the distribution of past data and discover irregularities data structure. This subsection literature review on the function of visualization and anomaly analysis in customer relationship management (CRM) and operation workflows. This visual representation of data patterns and operational bottlenecks can help financial institutions to improve on key area and make informed, data-driven decisions for their business. The following studies examine the relationship between these descriptive analytical techniques, and their real world uses in optimizing telemarketing strategies.

To begin with, Moro, Cortez and Rita (2014) provide a solid basis for analysing bank telemarketing data as they use descriptive analytics to identify customer segments. Unlike methodologies that apply a black-box machine learning algorithm to make the selections, their research present the importance of understanding the distribution of data over the past to make proper feature selections. For example, a certain job classifications and account balances have higher probability of term deposit subscriptions.

Follow up with the dynamic data interpretation, Turban, Sharda and Delen (2011) explore the role of visual analytics and the dashboard for turning raw descriptive data into actionable corporate strategies. Their research indicates that visual evidence such as clustered column charts and normalized stacked bar charts helps executives understand the business much quicker than raw spreadsheets and points to important business thresholds that raw spreadsheets don't reveal.

In the context of customer relationship management (CRM), Ngai, Xiu and Chau (2009) classify analytics applications from an anomaly-detection point of view. The study compares algorithmic outlier detection with visual anomaly detection, demonstrating that visual analysis can offer more immediate and interpretable context for business managers when they discover severe inefficiencies, such as extreme negative account balances or sudden increases in the number of calls.

Aksin, Armony, and Mehrotra (2007) explore operational metrics of call centres via resource allocation, workforce management, and aggregate volume of contacts. Their research compares volume driven operational models, which focus on making many calls per day with analytics driven targeted models, and shows that aggressively dialling for high call volume can sometimes be counterproductive and compromise employee productivity.

Finally, describe the essential enablers to get from the basic data mining to real data-driven decision-making (DDDM). The authors compare the purely technical model optimisation with value-driven model evaluation, which focuses more on the traditional accuracy scores and other scores. They argue that a model that is simpler to describe but still provides clear instruction to a specific business answer, is more useful to stakeholders than a more complex algorithmic model that is not interpretable.

2.5 Summary

This research analysed the literature and found that the application of business analytics and machine learning has greatly emerged in the banking sector as a tool used to assist in making evidence-based decisions. A lot of demographics, financial, transactional and campaign data are collected by financial institutions, and analysed to allow them to gain an understanding of customer behaviour, optimize their marketing efforts, and allocate organisational resources. Other research has shown repeatedly that

descriptive analytics, predictive modelling and customer analytics can assist banks to target the appropriate customer segments and optimize direct-marketing initiatives.

Research on business analytics applications highlighted the importance of data analysis, customer personalization and marketing optimization. Gul and Al-Faryan (2023) show how data analysis can help to make better decisions faster for managers, thus improving organisational productivity. Likewise, Torrens et al. (2022) demonstrated the potential of machine-learning platforms to analyse customer transaction data to detect sales opportunities and recommend personalised financial-product offerings. In addition, Nasir et al. (2024) demonstrated the ability of supervised machine-learning methods to help banks determine customers' potential to respond to promotional offers. Overall, these studies suggest that business analytics can be a valuable tool for enhancing customer satisfaction, improving marketing efficiency, and increasing the competitiveness of the organization.

Similar structured customer information and prediction problems were used in the research and demonstrated its suitability for supervised machine-learning applications. Hou et al. (2022) directly considered predicting customer subscription to bank term deposits, while Lemos et al. (2022), Singh et al. (2023) and Mouli et al. (2024) studied customer-churn prediction. Other researchers have used predictive analytics for loan default and credit-risk assessment. While these studies examined different business dimensions, they had some methodological similarities such as binary classification, customer-level data, categorical and numerical predictors, and imbalanced target variables.

Logistic Regression, Decision Tree, Random Forest, Support Vector Machine and XGBoost were the common machine-learning methods used in the reviewed studies. Logistic Regression was often used as a baseline model which was easy to understand, and Decision Tree models were used to give understandable non-linear decision rules. Generally, the Random Forest and XGBoost models performed well in prediction due to their ability to capture complex interactions among variables. However, it was not recognized in the literature that there was no optimal algorithm for everyone. Many aspects of the datasets, preprocessing choices, feature engineering, class balancing and evaluation criteria influenced the performance of the models.

The literature also points out that predictive models should not only be assessed by means of accuracy but also by other measures. Very often there are many more negative outcomes than positive ones in banking data sets. A model can thus be very accurate in predicting the majority class but miss out on identifying customers who have genuine subscription potential. Precision, recall, F1-score, ROC-AUC and PR-AUC, therefore, offer a more holistic evaluation of the prediction performance.

In a general way, the literature reviewed provides a solid background for the study at hand. But it also shows some disadvantages of class imbalance, data leakage, model interpretability, anomaly interpretation and business implementation of analytical insights. The present study aims at tackling these problems by integrating descriptive analytics, anomaly analysis and four machine-learning models in an integrated framework for business-oriented analytics.

2.6 Comparison

The 20 studies analysed in this research examined applications of business analytics, prediction problems in banking and customer-related use of analytics, machine-learning approaches, techniques for visualisation and approaches for making models interpretable. While the studies varied with regard to their research goals, data sources, data pre-processing, and evaluation measures, all of them had one goal: to convert customer data into information that can be used for organizational decision making.

The 20 studies examined are compared in Table 2.6.1 in terms of focus of the research, data context, analytical approach, findings and limitations.

No.	Author and Year	Research Focus	Dataset	Methods Used	Main Findings	Limitations
1	Gul and Al-Faryan (2023)	Data analytics and organisational productivity	Banking data	Business analytics	Analytics improved decision quality and organisational productivity	Limited focus on predictive campaign models
2	Torrens et al. (2022)	Personalised banking marketing	Customer transaction data	Machine-learning platform	Improved product recommendations and identification of sales opportunities	Implementation complexity
3	Nasir et al. (2024)	Marketing-response prediction	Banking campaign data	Supervised machine learning	Improved customer targeting and campaign effectiveness	Limited operational-cost analysis
4	Grassi et al. (2022)	Customer-data value creation	Open-finance data	Customer analytics	Improved personalisation and competitive advantage	Privacy and data-sharing concerns
5	Storey and Cohen (2002))	Cross-selling and campaign optimisation	Retail-banking data	Response modelling	Improved campaign profitability and resource allocation	Older modelling environment
6	Hou et al. (2022)	Term-deposit subscription	Banking-marketing data	Multiple machine-learning models	Machine learning effectively predicted customer subscriptions	Class-imbalance concerns
7	Lemos et al. (2022)	Bank customer churn	Brazilian financial data	Machine-learning classification	Customer behavioural information supported churn prediction	Target differs from term-deposit subscription
8	Singh et al. (2023)	Banking churn prediction	Retail-banking data	Machine learning and visualisation	Visual analytics improved the interpretation of churn predictions	Limited application to pre-call marketing decisions
9	Li et al. (2024)	Loan-default risk	Loan data	Multi-view Graph Convolutional Network	Improved loan-default risk prediction	Complex and less interpretable
10	Chen (2022)	Financial loan-default prediction	Financial customer data	Machine-learning models	Customer financial information supported default-risk prediction	Focused on credit risk rather than marketing
11	Mouli et al. (2024)	Customer-churn prediction	Telecommunications data	Logistic Regression, Decision Tree, Random Forest and XGBoost	Random Forest achieved the strongest performance	Non-banking dataset
12	Brito et al. (2024)	Retail-bank customer churn	More than three million customers and approximately 170 million transactions	XGBoost, Elastic Net, ADASYN and NearMiss	Advanced preprocessing and class balancing improved predictive performance	High implementation and computational complexity
13	Peng et al. (2023)	Churn prediction and model interpretation	Customer-churn data	GA-XGBoost and SHAP	Optimised XGBoost achieved strong performance, while SHAP improved interpretability	Computationally intensive
14	Chen (2024)	Credit-card customer churn	Credit-card customer data	Nine classification algorithms	XGBoost outperformed several alternative models	Limited discussion of business costs

15	Hon et al. (2023)	Bank customer churn	Multistate-bank data	Multiple machine-learning algorithms	SVM achieved the highest accuracy, while XGBoost produced strong F1-score and ROC-AUC results	Resampling methods such as SMOTE were not fully investigated
16	Moro et al. (2014)	Bank telemarketing success	UCI Bank Marketing dataset	Descriptive analytics, data mining and predictive modelling	Customer and campaign data supported term-deposit subscription prediction	Static profiling and possible data leakage from call duration
17	Turban et al. (2011)	Visual analytics and managerial decision-making	Organisational business data	Dashboards and interactive visualisation	Visual evidence improved managerial understanding and highlighted business thresholds	Greater focus on software and reporting than campaign strategy
18	Ngai et al. (2009)	CRM analytics and anomaly detection	Customer relationship management data	Data mining classification and visual anomaly analysis	Analytics helped identify customer patterns and operational irregularities	Anomalies were often treated as data-cleaning problems rather than business indicators
19	Aksin et al. (2007)	Call-centre operations and resource allocation	Call-centre operational data	Operations management and workload analysis	Targeted approaches improved resource use, while excessive calling produced diminishing returns	Limited integration of operational measures with customer demographic data
20	Provost and Fawcett (2013)	Data-driven decision-making and business value	General business data	Data mining and value-driven model evaluation	Interpretable analysis may provide greater business value than complex but unclear models	Limited direct application to bank telemarketing campaigns

Table 2.6.1: Comparison of Previous Studies

2.6.1 Comparison of Research Objectives and Datasets

The studies reviewed had multiple, but related, aims. Gul and Al-Faryan (2023), Torrens et al. (2022), Nasir et al. (2024), Grassi et al. (2022) and Storey and Cohen (2002), explored analytics as a comprehensive solution to enhance decision making, customer personalisation and marketing efficiency. Several of the studies studied the churn problem or the loan default risk, while Hou et al. (2022) and Moro et al. (2014) studied term-deposit subscription directly.

Although the predicted outcomes differed, most studies used structured demographic, financial, transactional and behavioural data to perform binary classification. The customers were categorized into subscribers, non-subscribers, active and churned customers, defaulting and non-defaulting borrowers. This resemblance indicates that supervised machine learning can be used in various banking decision problems.

The amount and availability of the datasets were also very variable. Reproducibility was achieved using public datasets; however, these yielded less organisation-specific information than proprietary datasets. In one case, Brito et al. (2024) worked with over three million customers and some 170 million transactions. Other researchers, however, had a more difficult task accessing and verifying private datasets.

2.6.2 Preprocessing vs Machine Learning methods

The typical approaches used in the studies were categorical encoding, numerical scaling, feature engineering, and feature selection and class balancing. Categorical customer variables were transformed using one-hot encoding and the numerical scaling was more important for Logistic Regression and Support Vector Machine.

The approaches used to deal with class imbalance varied among the studies. Some researchers used the original data distribution while others used SMOTE, ADASYN or NearMiss. Brito et al. (2024) showed that feature engineering and oversampling/undersampling can enhance the predictive performance. Re-sampleing can, however, lead to more false positives predictions and workload.

Logistic Regression was often employed as an interpretable baseline and Decision Tree gave interpretable non-linear rules. Random Forest was used to minimize the instability of a single tree by using multiple trees and it was found to be the best model by Mouli et al. (2024) in their study. Peng et al. (2023), Chen (2024) and Hon et al. (2023), however, found XGBoost to be very effective. The conflicting results of these studies clearly show that there is no single best algorithm. The result varies depending on the dataset, the choices made during preprocessing, the distribution of the classes and the evaluation metrics used.

2.6.3 Comparison of Evaluation and Interpretability

The accuracy was generally reported but not enough to evaluate imbalanced datasets. It is possible to have a model that was highly accurate by deciding that most people were not subscribers and not knowing that there was someone who actually was. More detailed studies, then, assessed precision, recall, F1-score and ROC-AUC.

Recall is crucial when the bank wants to maximise the number of potential subscribers that are identified, and precision is crucial when the bank wants to minimise unnecessary contacts. The F1-score is a trade-off of both. Another advantage of PR-AUC is that it focuses on performance of minority classes in an imbalanced dataset.

The reviewed studies also revealed that there is a trade-off between the performance and interpretability. Logistic Regression and Decision Tree were comparatively easy to explain while Random Forest, XGBoost and Graph Convolutional Networks were more complex. To tackle this issue, Peng et al. (2023) employed the SHAP to interpret the importance of the customer variables on the prediction of the XGBoost.

2.6.4 Comparison of Visualisation and Anomaly Analysis

For descriptive customer profiling, Moro et al. (2014) proved it could contribute to feature selection and elucidate the subscription behaviour. According to Turban et al. (2011) dashboards and interactive visualizations may be more effective at communicating business patterns than tabular reports. In a similar way, Singh et al. (2023) enhanced the interpretation of customer-churn predictions by leveraging visual analytics.

Ngai et al. (2009) emphasized the importance of analytics to uncovering unusual customer and operational behaviour. Anomalies were, however, treated mainly as data errors to be removed, however. This can fail to take into account significant aspects of the business, either in terms of the financial balance or the level of customer contact.

Aksin et al. (2007) also showed that call-centre strategies that are too aggressive may actually yield diminishing returns and make the operation less efficient. Provost and Fawcett (2013) highlighted that analytical recommendations and conclusions should not be evaluated solely based on technical performance but should be turned into real-life actions. All these studies help to support the incorporation of visual evidence, anomaly interpretation and business recommendations.

2.6.5 Overall Comparison

The literature reviewed highlights the potential to leverage business analytics for customer targeting, campaign management, customer retention, financial-risk assessment, etc. overall. While some ensemble methods like Random Forest and XGBoost were often more successful in predicting outcomes, other models like Logistic Regression and Decision Tree were still valuable due to their interpretability. It is also seen that the preprocessing methods, feature engineering and different class balancing methods and evaluation criteria affect model performance. A model that is very accurate may miss out on enough subscribers and a model that is very accurate may end up with too many false contacts. So the model that is most suited to the bank's campaign capacity, conversion goals and operating costs should be chosen.

Previous studies were also mostly focusing on predictive modelling, visualisation or anomaly detection separately. Very few that integrate descriptive analytics, interpretation of anomalies, predictive modelling and real business solutions in one. This restriction is the research gap described in Section 2.7.Note: Turban et al. (2011) and Provost and Fawcett (2013) are books, not peer-reviewed journal articles, so they are not included in the class. These are two journal studies and should be replaced with two relevant, peer reviewed journal articles, unless your lecturer states he or she will accept academic books.

2.7 Research Gap

The examined literature shows that machine-learning techniques are massively used in financial forecasting problems such as predicting term deposit subscription, customer attrition and loan default. Many previous studies, however, have just considered mean overall accuracy and it is possible for this to be misleading if the proportion of positive cases is small in the data set. A model can be very accurate if its prediction of a customer being a non-subscriber is accurate for most customers, but it is not accurate for those who are subscribers. Hence, there is a need for research that assesses the performance of models against minority classification metrics like Precision, Recall, F1-Score, ROC-AUC and PR-AUC. Another aspect of previous research was the lack of an analytical approach that combined descriptive analytics, anomaly analysis and predictive modelling within a unified business-oriented view.

Another drawback was that anomalies were generally regarded as technical data-cleaning problems. When a customer's account balance is at an extreme or the number of contacts to his or her campaign is high, the result of his or her previous campaigns, or the length of calls, may have significant insight into customer behaviour and operational efficiency. Automatically removing these observations could eliminate important patterns. Furthermore, the practical applicability of predictive models was not always taken into account. For instance, if a marketing call is made, duration is strongly associated with successful subscription, but is only available once the marketing call is made. Adding this variable can lead to over-optimized predictive accuracy but may not be of much help in targeting customers ahead of a call. Other studies failed to adequately discuss the effects of false positive and false negative predictions on the operation of the campaign, as these have a direct impact on the cost of the campaign, and the loss of subscription opportunities.

To fill these gaps, the present study combines descriptive analytics, anomaly analysis and predictive modelling for UCI Bank Marketing dataset. Four machine learning algorithms are compared under identical test conditions and two different training methods are compared: original and SMOTE balanced. To avoid data leakage and to make predictions based only on information prior to contact with the customer, the variable `duration' is omitted from the deployable predictive models. Accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC and confusion-matrix outcomes are used for assessing model performance. The final model recommendations are consequently technical and practical model recommendations for a campaign.

3.0 Problem Statement

Banks increasingly use direct marketing campaigns to promote their financial products. However, many campaigns apply the same strategy to their customer without considering their characteristics or purchasing behaviour. This often results in contacting uninterested customers, increasing marketing costs and reducing campaign effectiveness. The UCI Bank Marketing dataset also contains several data challenges, including class imbalance, “unknown” values, extreme outliers and excessive customer contacts. If these issues are not properly analysed, they may reduce model reliability and lead to ineffective business decisions. Although previous studies have used machine learning to predict customer responses, many focus mainly on accuracy and give limited attention to anomaly analysis, imbalanced-data metrics and practical business implications. Therefore, this study applies descriptive analytics, anomaly analysis and predictive modelling to the UCI Bank Marketing dataset. Logistic Regression, Decision Tree, Random Forest and XGBoost are compared using accuracy, precision, recall, F1-score and ROC-AUC to identify the most suitable model for improving customer targeting and marketing decision-making.

4.0 Anomaly Analysis & Solutions

To align with the initial data exploration phase, the raw dataset was scanned to identify any outliers, missing values, and unusual patterns in the data structure. These anomalies are It important to detect before the model is trained to avoid algorithmic bias and to make immediate operational changes.

4.1 Anomaly 1: Extreme Outliers in Customer Balance

4.1.1 Visual Evidence

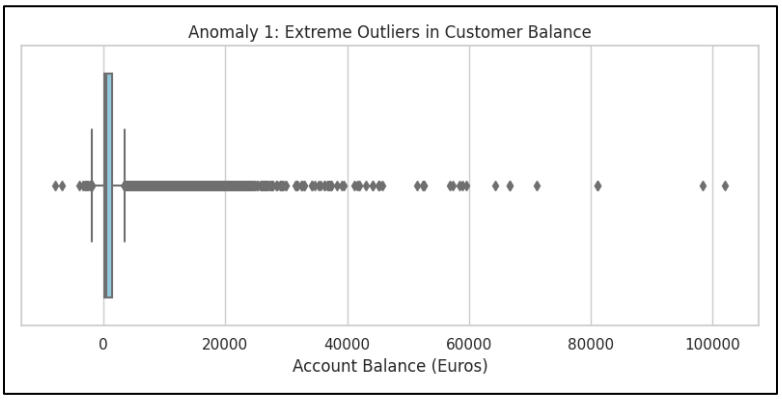


Table 4.1.1: Boxplot of Extreme Outliers in Customer Balance

A customer account balances audit shows that the interquartile range which includes most customers at the bank is tightly clustered around zero, while the tails are extremely abnormal. The data on the top end of the tail on the upper tale, exceeds enormously €100,000, and that's a small proportion of the populace that are not classifiable as traditional retail banking clients. On the other side of the coin, the lower part of the tail shows a concerning liquidity constraint, with invalid or extreme negative liquidity balances going down to -€8,000. These negative values are mostly unpaid debts or overdrafts. When people have negative liquidity and try to invest in a particular investment product, such as a term deposit, it doesn't make much economic sense because they don't have the basic capital that is required for investment.

4.1.2 Proposed Business Solution

The suggested business solution is to use standard, uniform telemarketing on such outlier financial situations, which is very inefficient. The bank needs to have an automated tier-based CRM routing system. One problem is that for the first time, accounts that have a negative balance should be removed from outbound call lists altogether, and this will instantly cut down on wasted labour costs and call centre fees on customers who, mathematically, cannot be converted. Secondly, it is important to automatically route balances of over €50,000 to dedicated wealth management advisors and away from regular call centre agents. This allows high-net-worth individuals to receive customized financial advice instead of a typical telemarketing sales pitch, which will help premium conversion rates.

4.2 Anomaly 2: Severe Over-contacting in Current Campaign

4.2.1 Visual Evidence

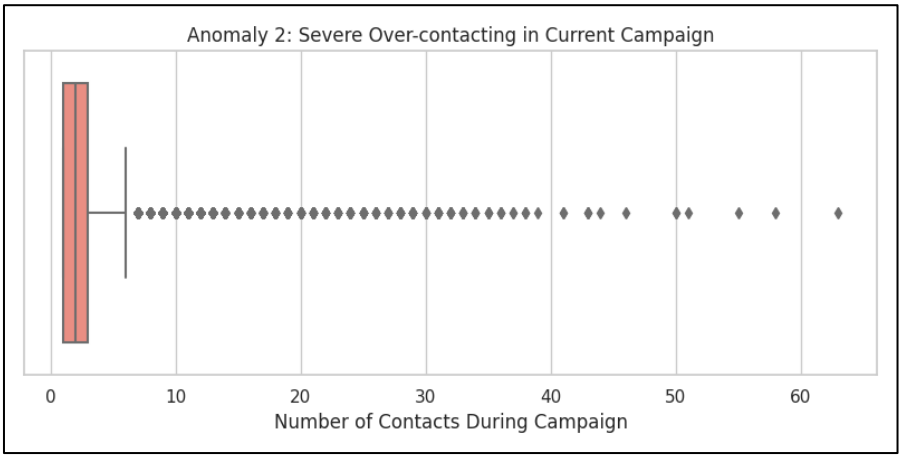


Table 4.2.1: Boxplot of Severe Over-contacting in Current Campaign

The visualization reveals a very unusual inefficiency in the way it is operating and the way that it is calling. The interquartile range shows that most customers are being contacted between 1 and 3 times, but a long tail of extreme statistical outliers exists. The data shows that there is some people who got call between 40-63 times in a single campaign, but they never subscribed. These are not errors of data entry, but instead a legitimate observation of the business with a very aggressive call volume approach. Over contacting indicates a failure in campaign management, demonstrating high levels of wasted employees as well as potential serious damage to the brand and causing customer nuisance.

4.2.2 Proposed Business Solution

The suggested Business Solution is a hard CRM system cap, e.g. only 5 contacts per campaign shall be allowed, which must be implemented immediately by the management. When a customer reaches this point, the system could automatically put a "fatigued" tag on them and move them back to the "dialling" pipeline so that those labour hours can be shifted to new customers.

4.3 Anomaly 3: Massive Missing Data Structure

4.3.1 Visual Evidence

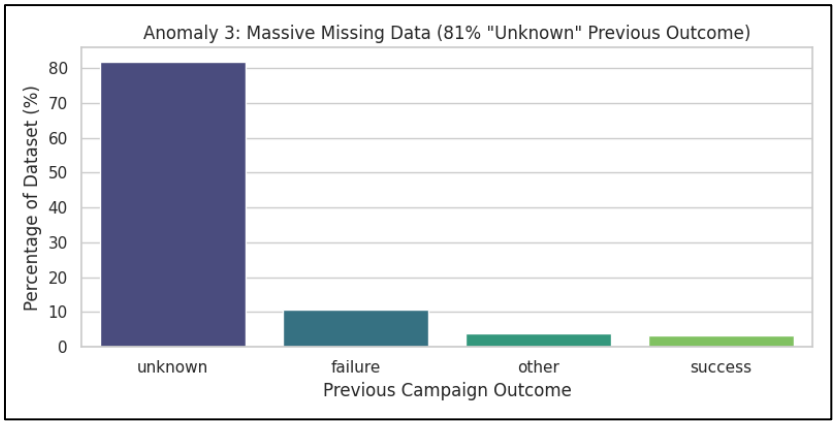


Table 4.3.1: Bar Chat of Missing Data

Overall, more than 81 percent of the records in the dataset report an outcome in the previous campaign as "unknown. In the conventional statistical analytics, such a missing data rate of 81% would be considered too high for the column to be included or the data would be treated with such mean/mode substitution that the usefulness of the column would greatly be diminished. But in this particular scenario, it is a structural error, not a data collection error. It's a business fact that a majority of the existing contact list is new leads or cold leads who have not been a part of any previous marketing campaign. Thus, substituting "unknown" with the most common known category (such as "failure") would artificially create customer history and significantly misrepresent the model's perception of customer lead maturity.

4.3.2 Proposed Business Solution

One of the proposed business solutions is that the marketing department is to cease to treat the customers as a monolithic body. The business approach is to divide the operational workflow and create two different telemarketing scripts. The first one is an "Acquisition Script" specifically designed for the 81% "unknown" audience, with emphasis on brand introduction and low barrier product education. The second type should be a "Retention and Cross-sell Script" for those 19% of customers who have a history of engaging with you or not, using their past data. Analytically, this makes sense of the data pre-processing technique of keeping the "unknown" variable as an independent categorical variable, which gets one-hot encoded, so that the algorithms can explicitly model the action of new customers compared to the old ones.

4.4 Anomaly 4: Structural Anomaly in Days Passed (pdays)

4.4.1 Visual Evidence

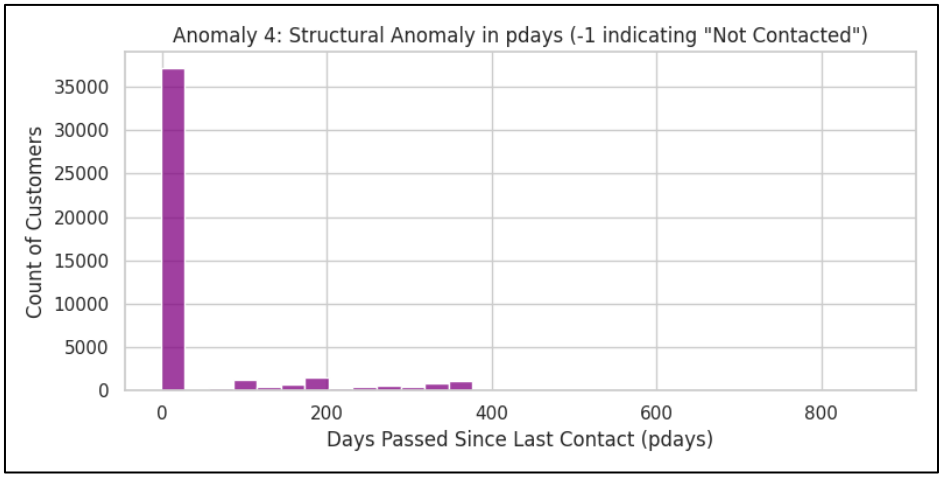


Table 4.4.1: Bar Chat of Structural Anomaly in pdays

There is a heavy and very conspicuous mass of -1s for the pdays variable, which is not natural. In a mathematical sense this results in a rather severe left-skew. This does not mean it is an impossible negative time value or a glitch in the system but rather it is a system generated structural anomaly representing "Not Previously Contacted. It is a categorical value that is stored as an integer but passed as arguments to distance-based algorithms which would incorrectly assume that -1 was "closer" to 0 days than 100 days.

4.4.2 Proposed Business Solution

What is proposed as a business solution is that from a business operation standpoint, this large group of minus 1, means that the bank's marketing strategy is too focused on cold calling completely new prospects versus working on warmer leads that have had a history of engagement. The company should invest some of its marketing dollars into creating follow-up campaigns to boost conversions, as hot leads tend to convert at a higher rate, which results in a higher ROI. One of the new binary indicators features such as is_previously_contacted: 1 or 0 needs to be built analytically. This means that predictive algorithms can assume the -1 is not a negative measurement of time, but simply a categorical state of being a new lead.

4.5 Anomaly 5: Extreme Call Durations

4.5.1 Visual Evidence

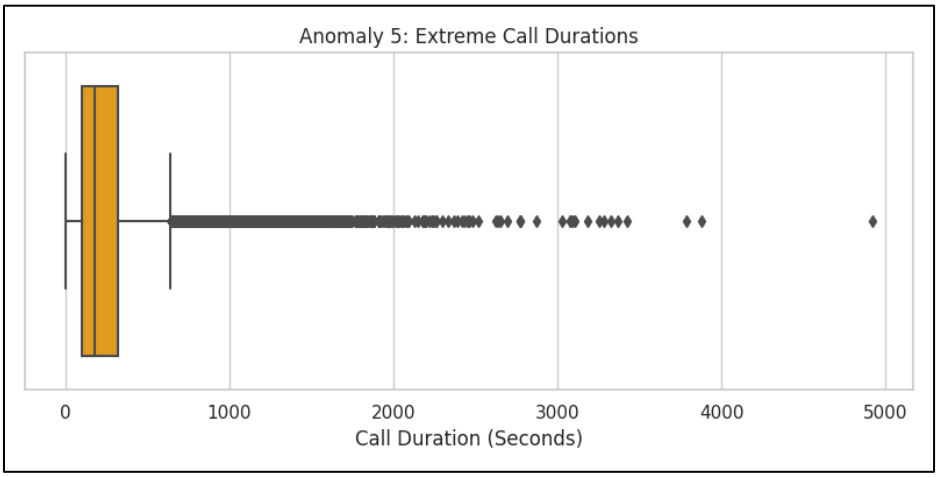


Figure 4.5.1: A boxplot capturing the length of the telemarketing calls in seconds.

The Extreme Call Durations (duration) is a type of anomaly that occurs when calls are either too long or too short. The data shows call lengths that are far longer than standard telemarketing call times, with some calls exceeding 5,000 seconds, which is just over 80 minutes. An 80-minute call is a major constraint on a typical retail banking term deposit sell.
These additional exchanges disrupt the call centre timetable which affect Average Handle Time (AHT) statistic and contribute to the waiting time for other potential customers. However, one of the biggest risks to the analytics pipeline is duration as call's duration is closely related to the success of a subscription but can be unknown at time of call. Therefore, including it to a predictive model provides an opportunity for the model to “cheat” by making a prediction based on the post-event metrics.

4.5.2 Proposed Business Solution

Call centre managers need to set tight handle-time limits operationally, and it is proposed that these are aligned with the Business Solution. Call centre managers should be forced to adhere to rigid handle-time requirements operationally and it is suggested that these be aligned with the Business Solution. The agents should be instructed to end the call if it's running too long or to create a follow-up appointment. From an analytical point of view, this discrepancy determines a mandatory preprocessing step. In all major predictive models, the duration column was explicitly excluded as detailed in Section 5.2.3. This ensures that the machine learning algorithms produce probability scores using only pre-call demographic and historical data, so that the result is an actual deployable predictive model in a real-world, real-time targeting environment.

5.0 Methodology

5.1 Dataset Description

Variable	Description
age	Customer age.
job	Type of job.
marital	Marital status.
education	Education level.
default	Whether the customer has credit in default.
balance	Customer account balance.
housing	Whether the customer has a housing loan.
loan	Whether the customer has a personal loan.
contact	Contact communication type.
day	Day of the last contact.
month	Month of the last contact.
duration	Last contact duration, in seconds
campaign	Number of contacts performed during the current campaign.
pdays	Number of days since the customer was last contacted in a previous campaign.
previous	Number of contacts performed before the current campaign.
poutcome	Outcome of the previous marketing campaign.
y	Whether the customer subscribed to a term deposit.

Table 5.1: Dataset Descriptions

5.2 Data Preprocessing

Check	Evidence	Decision
Dataset dimensions	45,211 records and 17 variables	Retain the complete source dataset before splitting
Duplicate rows	0 detected	No duplicate removal required
Null values	0 across all variables	No numerical or categorical imputation required
Blank strings	0 across categorical fields and target	No blank-string replacement required
Target imbalance	39,922 no (88.30%); 5,289 yes (11.70%)	Use stratification and minority-class metrics; test SMOTE on training data only
pdays = -1	Defined as not previously contacted	Retain as valid business information
duration	Known only after the call	Retain for descriptive analysis but exclude from deployable pre-call models

Table 5.2.1: Raw-data audit and final preprocessing decisions

The predictive workflow started with the repeatable audit of the entire Bank Marketing data set. The working file had 45,211 customer-contact records and 17 variables, consisting of 16 predictors and the binary target y. No duplicated rows, null values or blank strings were found in the source data during the audit. As a result, no observation has been discarded simply to produce an artificially clean

modelling data set. In contrast, each abnormal value was evaluated with regard to its business significance prior to making a decision about preprocessing.

5.2.1 Treatment of "unknown" Categories

Variable	Unknown records	Share of dataset	Final treatment
poutcome	36,959	81.75%	Retained as a separate category
contact	13,020	28.80%	Retained as a separate category
education	1,857	4.11%	Retained as a separate category
job	288	0.64%	Retained as a separate category

Table 5.2.2: Frequency and treatment of "unknown" categories

The text value "unknown" was present in selected categorical variables, particularly poutcome and contact. These entries were not considered as "normal" missing cases because they occur in the campaign records as an observed state. Their deletion would eliminate a substantial amount of the data, and their replacement with the most common category would result in unsupported information. They also had different subscription rates than the groups they were known to be subscribing to, suggesting that "unknown" contained predictive information. As a result, each of these missing values was represented separately as a categorical level and dealt with using one-hot encoding.

5.2.2 Numerical Outliers and Validity Checks

Variable	Count	Share	Lower bound	Upper bound	Decision
previous	8,257	18.26%	0.0	0.0	Retained; reviewed as valid business observations
pdays	8,257	18.26%	-1.0	-1.0	Retained; reviewed as valid business observations
balance	4,729	10.46%	-1962.0	3462.0	Retained; reviewed as valid business observations
duration	3,235	7.16%	-221.0	643.0	Retained; reviewed as valid business observations
campaign	3,064	6.78%	-2.0	6.0	Retained; reviewed as valid business observations
age	487	1.08%	10.5	70.5	Retained; reviewed as valid business observations
day	0	0.00%	-11.5	40.5	Retained; reviewed as valid business observations

Table 5.2.3: IQR-based outlier audit

The interquartile-range rule identified statistically extreme values in balance, duration, campaign, age, pdays and previous. These were not automatically deleted. The negative balances are legitimate, as are repeated contacts and pdays that are very large or very small, as customers who have not been in contact with a campaign appear in the concentration at 0. The validity checks also did not detect impossible ages, invalid calendar days, negative call durations, campaign values of less than 1, or pdays values of less than -1 or negative previous-contact counts. The modelling data was thus preserved with genuine extremes, and tree-based algorithms were therefore given the opportunity to learn non-linear patterns from these extremes. Scaling was used only where required by Logistic Regression.

5.2.3 Target Encoding, Class Imbalance and Leakage Control

Duration group	Customers	Subscription rate
0–30	2,010	0.25%
31–60	2,756	0.15%
61–120	9,278	2.18%
121–300	18,893	8.57%
301–600	8,484	19.15%
More than 600	3,790	48.36%

Table 5.2.4: Duration groups and observed subscription rates

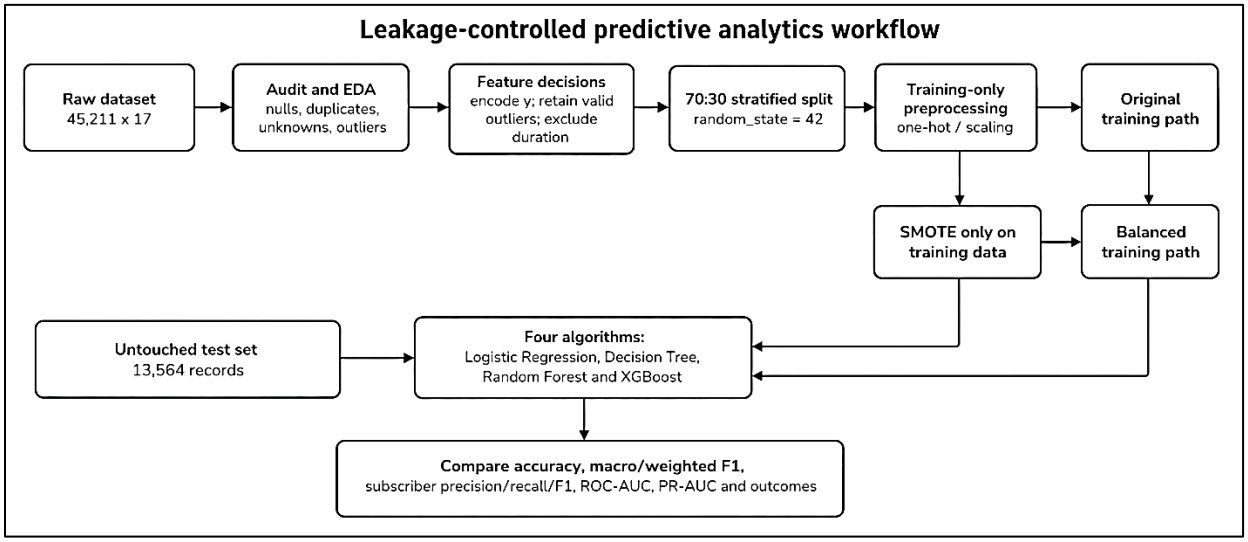


Figure 5.2.1: Leakage-controlled predictive analytics workflow

The target y was encoded as 0 for no subscription and 1 for subscribed, making the minority class the positive class throughout model training and evaluation. The ratio of the class was about 7.55 non-subscribers to one subscriber. This distribution was preserved by splitting the data in a stratified fashion with a ratio of 70:30 and a random_state = 42. The split resulted in 31,647 training records and 13,564 testing records. The training target had 27,945 class-0 records and 3,702 class-1 records, and the testing target had 11,977 class-0 records and 1,587 class-1 records.

The predictor duration was excluded from all main predictive models. Duration is strongly associated with subscription, its value becomes known only after the end of a telephone call. For instance, calls between 600 and 900 seconds had a 48.36% subscription rate, whereas 0–30 seconds had a 0.25% rate. Including duration would therefore over-inflate performance without justifying the bank's pre-call targeting decision. The variable was measured but not used for splitting X and y.

After duration and y were excluded from the predictor matrix, the modelling dataset contained 15 predictors. The categories variables were one-hotted with the option handle_unknown set to ignore so that the fitted transformer could handle unknown categories that may only be seen at test time. Numerical predictors were standardised only for Logistic Regression because coefficient-based optimisation is sensitive to differences in scale. Environmental variables were retained in numeric form in Decision Tree, Random Forest and XGBoost since they do not split based on distance.

All preprocessing objects were fitted using X_train only and then applied to X_test. This ensured that no information from the test set was carried over to the levels of the categories, scaling, or model fitting. SMOTE was used only after the training predictors were transformed for the balanced model

versions. This minority training class was expanded from 3,702 to 27,945 synthetic and original positive observations, resulting in a training set of 55,890 observations. No re-sampling of the test set took place, and it was representative of the distribution from the initial campaign.

5.3 Descriptive Analytics Methodology

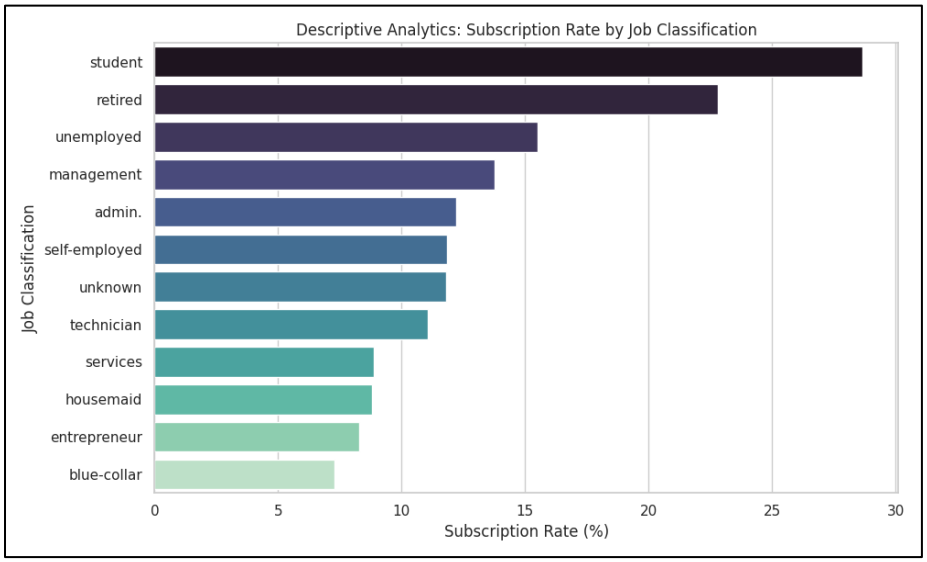
Following the data preprocessing phase, descriptive analytics was performed on the cleaned dataset of 45,211 customer records, which finished the predictor matrix, and removed some variables that leaked information, such as call duration. The primary objective of this phase was to define foundational demographic baselines parameters to identify univariate statistical trends and detect the non-linear relationships between independent features and the encoded target variable (y) in the labels prior to algorithmic training.

5.3.1 Summary Statistics & Percentages

Variable	Count	Mean	Median	Std Dev	Min	Q1 (25%)	Q3 (75%)	Max	Skewness
Age	45,211	40.94	39.00	10.62	18.00	33.00	48.00	95.00	+0.68
Balance (€)	45,211	1,362.27	448.00	3,044.77	-8,019.00	72.00	1,428.00	102,127.00	+8.36
Campaign	45,211	2.76	2.00	3.10	1.00	1.00	3.00	63.00	+4.90
Pdays	45,211	40.20	-1.00	100.13	-1.00	-1.00	-1.00	871.00	+2.62
Previous	45,211	0.58	0.00	2.30	0.00	0.00	0.00	275.00	+41.85

The initial descriptive methodology involved using Python's Pandas library to generate central tendency and dispersion metrics such as mean, median, standard deviation, min, max. These are applied for continuous variables such as age, balance, and campaign. Moreover, frequency distributions were used for categorical variables such as job, marital, education were evaluated using percentages. This step helps to understand the post-split demographic shape of the target audience and confirm that the majority class of the bank's dataset married people with secondary education in standard employment.

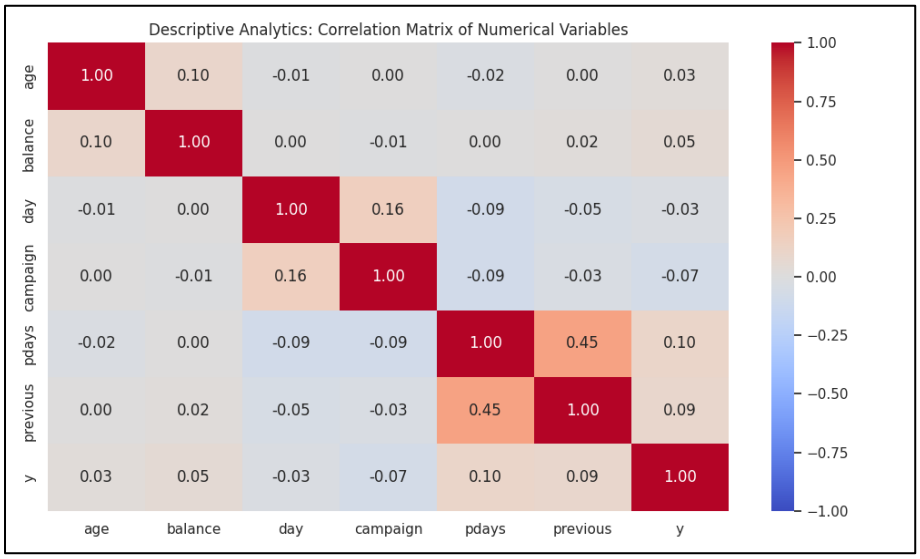
5.3.2 Cross-Tabulation & Target Mapping



To directly link demographic traits to business value, cross-tabulation (bivariate analysis) was performed. This approach summed the binary target variable which is y = 1 across different categorical segments to calculate the average subscription rate and revealed important differences in behaviour between the bank's labour and social classes.

One of the findings is although blue collar (9732 records) has the most occupational group, their conversion rate for this group is one of the lowest at 7.2%. In contrast, cross-tabulation successfully identify the non-active labour segments as the highest-rewarding targets. The leading group in terms of conversion rate is students (28.6%), followed by retirees (22.7%). Similarly, single individuals are more likely to convert (14.9%) than married people (10.1%), and those with tertiary education more likely to convert (15.0%) than those with secondary education (10.5%). This mapping shows that there is no direct correlation between raw customer volume and campaign success.

5.3.3 Correlation Analysis & Multicollinearity Checks



Pearson correlation matrix was applied across all continuous numerical variables in the cleaned data set to ensure that the following predictive modelling phase was robust. The mathematical check performed two vital roles in the preparation of the model which are verification of independence and predictive baseline assessment.

On verification of independence, multicollinearity variables were not severe as indicated by the matrix among the main demographic. For example, age and balance have a weak correlation of 0.10, stating that there is no variance inflation when using linear baseline models like Logistic Regression.

On predictive baseline assessment, a moderate positive correlation (0.45) was found between pdays and previous, which is a logical correlation as both metrics are measuring historical campaign engagement. Moreover, it showed that other pre-call variables have near zero linear correlation with the target variable y. The non-linearity of the data suggests that the use of advanced ensemble methods that are non-linear is mathematically justified in Section 5.4, as simple linear models cannot accurately classify the data due to its complex nature and hidden thresholds. The models for example are such Random Forest and XGBoost.

5.4 Predictive Analytics: Methodology

Model	Final configuration	Preprocessing	Role in comparison
Logistic Regression	C = 1.0; liblinear; max_iter = 2,000	StandardScaler + one-hot encoding	Interpretable baseline; coefficients show direction of predictive association
Decision Tree	Grid search; gini; max_depth = 10; min_samples_leaf = 20; min_samples_split = 50	One-hot encoding; no scaling	Transparent non-linear decision rules; depth and leaf controls reduce overfitting
Random Forest	Randomized search; 200 trees. Original: depth 16, min split 10, leaf 1. SMOTE: depth 12, min split 2, leaf 5	One-hot encoding; no scaling	Ensemble reduces variance and captures interactions; tuned for subscriber F1
XGBoost	200 trees; depth 5; learning rate 0.05; subsample 0.8; column sample 0.8	One-hot encoding; no scaling	Strong boosted-tree probability ranking; settings aligned with the lecturer sample workflow

Table 5.4.1. Predictive models, configurations and justification

The predictive task was rephrased as a binary classification: predict if a customer would take on a term deposit before the bank would make a marketing call. Four algorithms were chosen to offer complementary modelling perspectives. The Logistic Regression delivered an easily understood linear baseline, the Decision Tree gave easily understood non-linear rules, the Random Forest mitigated instability of the single tree by averaging over many, and XGBoost was a sequential boosting model that captured complex interactions. Each algorithm was evaluated using both the original imbalanced training data and a SMOTE-balanced version, producing eight comparable model variants.

5.4.1 Validation and Hyperparameter Selection

All eight model variants used the same training and testing observations. Decision Tree and Random Forest were tuned with stratified cross-validation using subscriber F1-score as the optimisation measure. Decision Tree performed a five-fold grid search over criterion, depth and minimum-node-size. Random Forest used a randomized search across the number of trees, depth, feature sampling and minimum-node-size settings. Logistic Regression and XGBoost used fixed reproducible settings aligned with the lecturer's sample workflow; these models were not described as exhaustively tuned. Every fitted pipeline used random_state = 42 where the algorithm supported it.

5.4.2 Evaluation Measures

Measure	Technical meaning	Business meaning
Testing accuracy	Overall share of correct predictions	Useful only with minority-class measures because the data are imbalanced
Subscriber precision	TP / (TP + FP)	Expected conversion quality among customers selected for priority contact
Subscriber recall	TP / (TP + FN)	Share of all potential subscribers captured by the model

Subscriber F1	Harmonic mean of precision and recall	Primary balanced-classification selection measure
Macro F1	Average F1 across both classes	Checks whether performance is balanced rather than dominated by non-subscribers
ROC-AUC	Ranking discrimination across thresholds	Measures how well probabilities separate subscribers from non-subscribers
PR-AUC	Precision-recall performance across thresholds	Preferred ranking measure when the positive class is uncommon

Table 5.4.2:Evaluation measures and business interpretation

The accuracy was reported for completeness but not used independently, because a classifier that classifies all customers as non-subscribers would already obtain the accuracy of about 88.30%. The main measures were subscriber precision, subscriber recall and subscriber F1-score. Precision estimates the percentage of the selected customers that actually subscribed, while recall is the percentage of all actual subscribers captured, and F1 is a compromise between both. Macro F1 was balanced by giving equal weights to the two classes, and weighted F1 was biased by class prevalence. ROC-AUC was calculated to evaluate overall discrimination at all thresholds, and Precision–Recall AUC was highlighted as it directly measures the ranking of positive classes in an imbalanced data set. Confusion-matrix counts were also translated into true-positive subscribers, unnecessary false-positive contacts and missed subscribers.

5.4.3 Reproducibility Controls

- The same stratified 70:30 split and unchanged test set were used for all models.
- One-hot encoders and scalers were fitted only on training data.
- SMOTE was applied only to transformed training data and never to testing data.
- The positive class was consistently defined as subscription = 1.
- All model outputs were saved as metric tables, predictions, summaries and fitted model files.
- The final comparison notebook recalculated every metric from the prediction files; the maximum difference from saved results was 0.0000.

6.0 Results and Discussion

6.1 Predictive Results

The eight model variants were evaluated on the same 13,564-record test set containing 1,587 subscribers and 11,977 non-subscribers. The final comparison notebook independently recalculated all metrics from the saved prediction files and confirmed an absolute difference of zero against the model-specific summaries. This verification supports a fair comparison under identical test conditions.

Model	Accuracy	Subscriber precision	Subscriber recall	Subscriber F1	ROC-AUC	PR-AUC
Logistic Regression - Original	89.44%	68.08%	18.27%	28.81%	76.97%	41.88%
Logistic Regression - SMOTE	74.75%	25.93%	62.38%	36.63%	76.45%	41.38%
Decision Tree - Original	89.37%	61.35%	24.70%	35.22%	74.02%	36.53%
Decision Tree - SMOTE	88.70%	52.30%	38.63%	44.44%	76.24%	37.87%
Random Forest - Original	89.62%	65.09%	24.32%	35.41%	79.66%	46.27%
Random Forest - SMOTE	89.09%	54.85%	38.12%	44.98%	79.02%	44.31%
XGBoost - Original	89.64%	65.58%	24.13%	35.28%	80.31%	46.87%
XGBoost - SMOTE	89.61%	66.54%	22.56%	33.69%	79.86%	45.55%

Table 6.1.1. Final predictive performance on the common test set

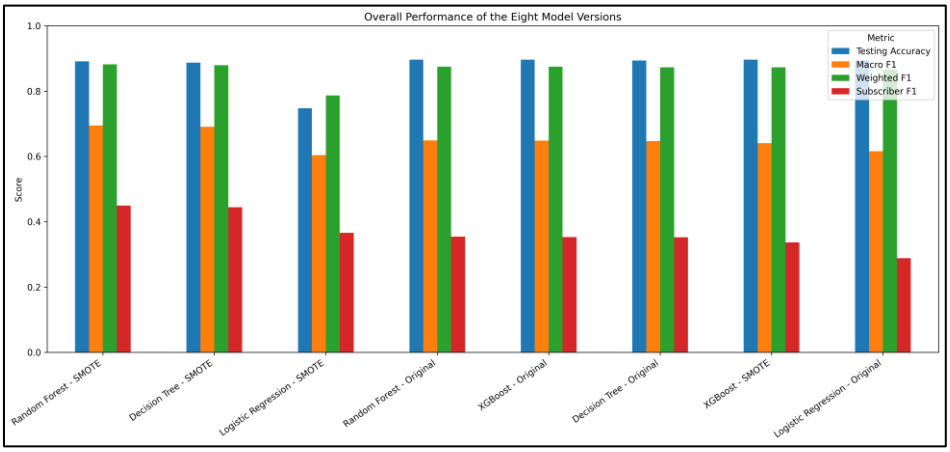


Figure 6.1.1. Overall performance of the eight model versions

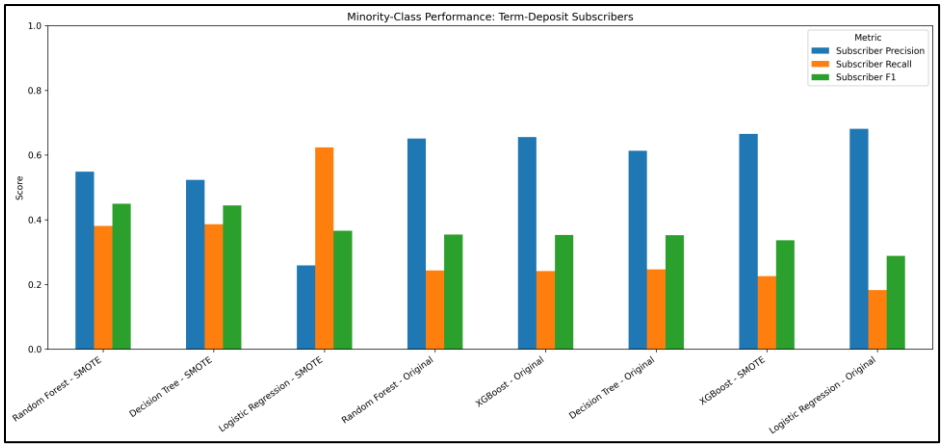


Figure 6.1.2. Minority-class precision, recall and F1 performance

6.1.1 Logistic Regression Results

The original Logistic Regression model achieved 89.44% accuracy and the highest subscriber precision of all variants at 68.08%, but its recall was only 18.27%. It correctly identified 290 subscribers and missed 1,297. After SMOTE, recall increased sharply to 62.38%, the highest recall in the study, and true positives increased to 990. However, precision fell to 25.93%, accuracy fell to 74.75%, and false-positive contacts increased to 2,828. The result shows that SMOTE shifted the linear decision boundary towards subscriber detection, but at a substantial operational cost in unnecessary calls.

6.1.2 Decision Tree Results

The original Decision Tree achieved 89.37% accuracy, 61.35% subscriber precision, 24.70% recall and a 35.22% subscriber F1-score. The SMOTE version improved recall to 38.63% and subscriber F1 to 44.44%, while precision remained moderate at 52.30%. It identified 613 subscribers and generated 559 false-positive contacts. Training and testing accuracy were close in both tree variants, suggesting that the selected depth and minimum-node-size controls limited severe overfitting.

6.1.3 Random Forest Results

The original Random Forest achieved 89.62% accuracy, 65.09% subscriber precision, 24.32% recall, 35.41% subscriber F1, 79.66% ROC-AUC and 46.27% PR-AUC. The SMOTE variant reduced precision to 54.85% but increased recall to 38.12%, producing the highest subscriber F1-score in the study at 44.98%. It correctly identified 605 subscribers, missed 982 and generated 498 false-positive contacts. Its cross-validation F1 of 43.19% was close to its test F1 of 44.98%, supporting reasonable stability beyond the training sample.

6.1.4 XGBoost Results

Original XGBoost produced the highest testing accuracy at 89.64%, the highest ROC-AUC at 80.31% and the highest PR-AUC at 46.87%. Its subscriber precision was 65.58%, but recall remained low at 24.13%, giving a subscriber F1-score of 35.28%. The SMOTE version did not improve XGBoost: subscriber recall declined to 22.56% and F1 declined to 33.69%. Therefore, the original XGBoost variant was retained as the stronger probability-ranking model.

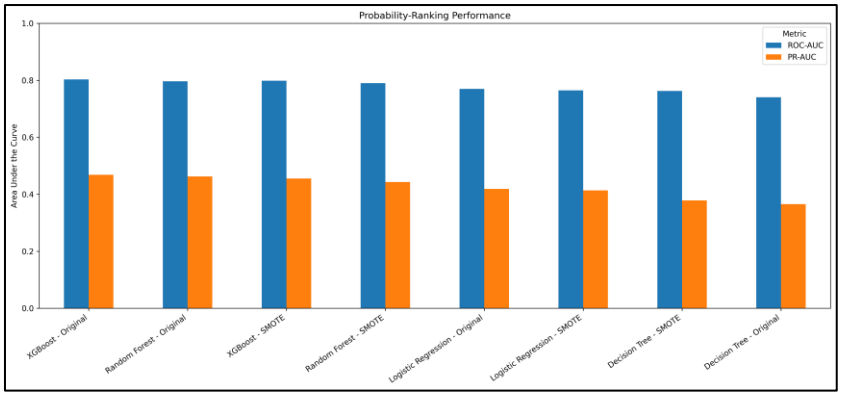


Figure 6.1.3. Probability-ranking performance across all model variants

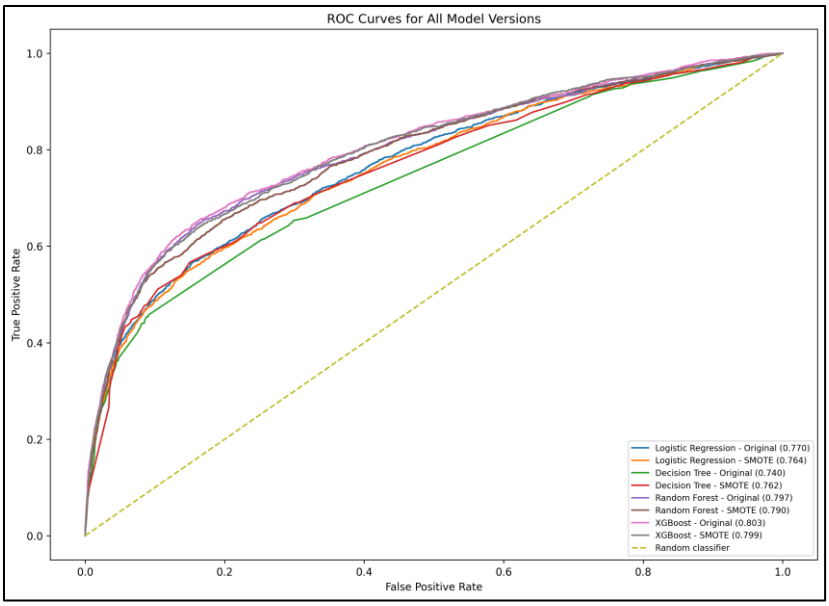


Figure 6.1.4. Combined ROC curves for all model variants

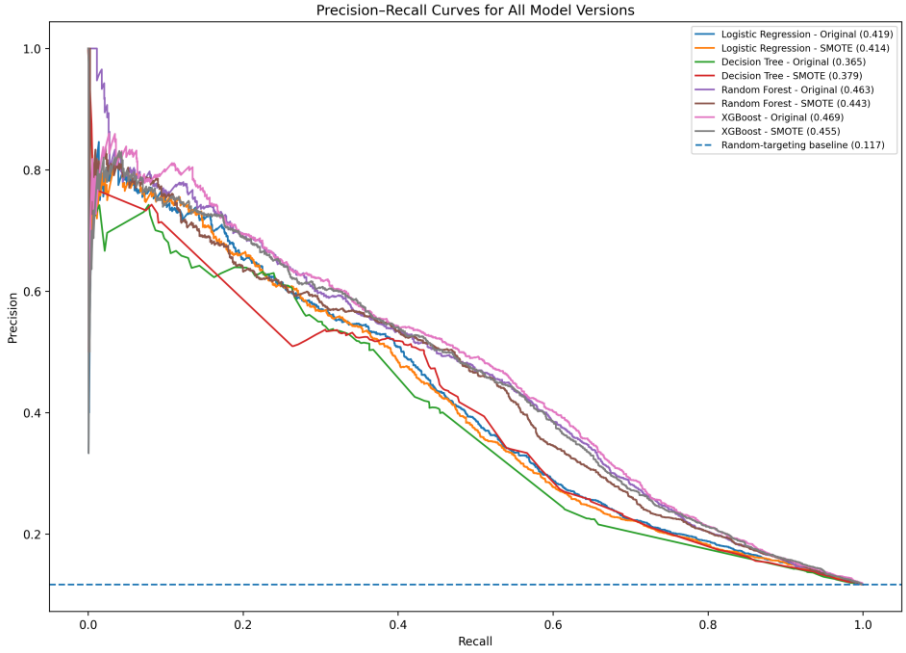


Figure 6.1.5. Combined Precision–Recall curves for all model variants

6.1.5 Confusion-Matrix Outcomes

The default classification threshold of 0.50 produced materially different campaign workloads. Precision-oriented original models selected fewer customers and generated fewer false positives, whereas the balanced models generally captured more subscribers. The SMOTE Logistic Regression produced the greatest subscriber coverage but selected 3,818 customers and created 2,828 false-positive contacts. Random Forest with SMOTE offered a more controlled trade-off, selecting 1,103 customers, identifying 605 genuine subscribers and generating 498 false positives.

Model	True-positive subscribers	False-positive contacts	Missed subscribers	Customers selected
Logistic Regression - Original	290	136	1,297	426
Logistic Regression - SMOTE	990	2,828	597	3,818
Decision Tree - Original	392	247	1,195	639
Decision Tree - SMOTE	613	559	974	1,172
Random Forest - Original	386	207	1,201	593
Random Forest - SMOTE	605	498	982	1,103
XGBoost - Original	383	201	1,204	584
XGBoost - SMOTE	358	180	1,229	538

Table 6.1.2. Campaign outcomes at the default 0.50 threshold

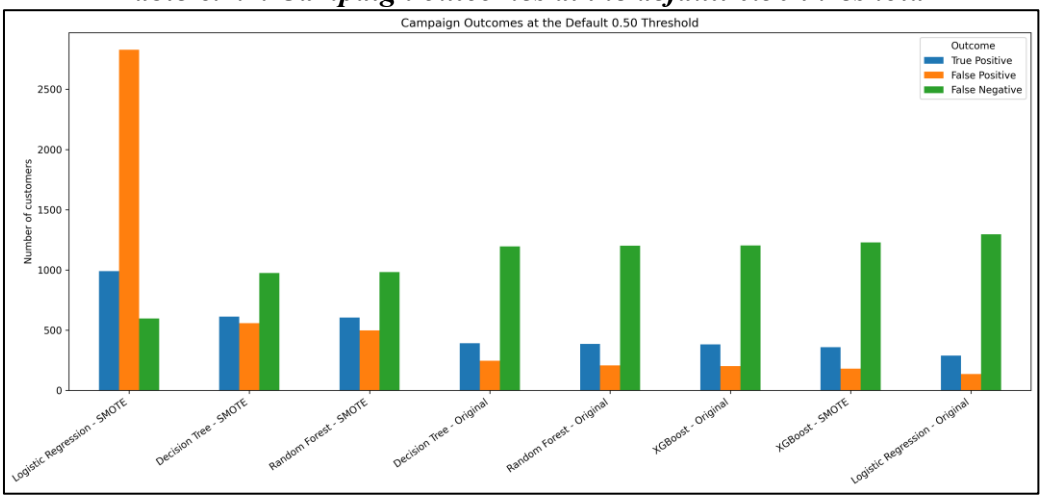


Figure 6.1.6. Campaign outcomes at the default 0.50 threshold

6.1.6 Coefficients and Feature Importance

The models identified several recurring predictive signals. A successful previous campaign outcome was the strongest positive association in Logistic Regression and the most important feature in the original Decision Tree, Random Forest and XGBoost models. Contact channel, housing-loan status, campaign month, age, balance and prior-contact history also influenced predictions. Logistic Regression associated March, September, December and October contacts, as well as retired and student occupations, with higher subscription likelihood. Negative coefficients were observed for unknown contact method, January, November and August contacts, previous-campaign failure and

housing-loan status. Feature importance and coefficients describe predictive association within each model; they do not establish causal effects.

Model evidence	Highest-ranked signals	Interpretation caution
Logistic Regression - Original	poutcome_success; month_mar; month_sep; month_dec; month_oct	month_jan; contact_unknown; month_nov; month_aug; poutcome_failure
Decision Tree - SMOTE	housing_yes; contact_cellular; poutcome_success; marital_single; month_aug	Tree importance has no positive/negative direction
Random Forest - SMOTE	housing status; contact_cellular; poutcome_success; month_aug; marital status	Importance ranks contribution but not direction
XGBoost - Original	poutcome_success; contact_unknown; housing_yes; month_mar; month_apr	Importance ranks contribution but not direction

Table 6.1.3. Main predictive signals from coefficients and feature importance

6.2 Model Comparison & Recommendation

Business priority	Recommended model	Selection measure	Score
Limited calling capacity and high conversion quality	Logistic Regression - Original	Subscriber Precision	68.08%
Capture as many potential subscribers as possible	Logistic Regression - SMOTE	Subscriber Recall	62.38%
Balance subscriber precision and recall	Random Forest - SMOTE	Subscriber F1	44.98%
Rank promising customers in an imbalanced campaign	XGBoost - Original	PR-AUC	46.87%
Overall probability discrimination	XGBoost - Original	ROC-AUC	80.31%

Table 6.2.1. Best model for each business priority

The final model was chosen based on the decision objective and not just a single headline score. Accuracy was not considered to be a valid metric, since it was only marginally higher than the 88.30% majority class baseline for all of the models that were concerned with precision. The comparison focused on the following: subscriber F1 for balanced classification; probability ranking (PR-AUC); interpretability; operational cost of false-positive calls and the loss of subscribers; and stability of the cross validation.

6.2.1 Primary Classification Recommendation: Random Forest with SMOTE

Random Forest with SMOTE is the suggested primary classification model since it had the highest PR-AUC of 44.31%, the highest macro F1 score of 69.46% and the highest subscriber F1 score of 44.98%. At the default threshold it correctly identified 605 subscribers, missed 982 and generated 498 false-positive contacts. Random Forest with SMOTE yielded 61 fewer false positives and a slightly better F1 score, despite being able to detect eight more subscribers than Decision Tree with

SMOTE. It also had an F1 of 43.19% in cross validation, with the test F1 of 42.19%, meaning that the chosen model was not overly reliant on a positive test result.

Operationally, this model provides the strongest observed balance between conversion quality and subscriber coverage. The proportion of actual subscribers selected by the model was 54.85 per cent of all the customers, whereas the natural test-set subscription rate was 11.70 per cent. This is a considerable targeting lift without the extremely high call volume of SMOTE Logistic Regression.

6.2.2 Alternative Ranking Recommendation: Original XGBoost

Original XGBoost was recommended as the alternative customer-ranking model because it is the best model in terms of ROC-AUC (80.31%) and PR-AUC (46.87%). These results show that its probability scores were the best overall ranking of customers by their likelihood to subscribe. It has a default-threshold recall of only 24.13%, thus it should not be used in place of Random Forest for the goal of balanced classification at a fixed threshold of 0.50. However, it is suitable for ranking customers and selecting the highest-probability segment according to a limited daily or weekly calling capacity.

6.2.3 Why the Other Models Were Not Selected as the Primary Model

Original Logistic Regression had the highest subscriber precision at 68.08% which is appealing for a bank that can follow up with only a very few of the most confident prospects. Nevertheless, it missed 81.73% of actual subscribers. SMOTE Logistic Regression achieved the highest recall at 62.38%, but its precision of 25.93% and 2,828 false positives would create a much larger calling workload. Decision Tree with SMOTE provided competitive F1 and slightly higher recall than Random Forest with SMOTE, but Random Forest reduced false positives and achieved stronger ROC-AUC and PR-AUC. The original Random Forest and both the XGBoost models ranked customers well, but were not able to “catch” enough subscribers at the default threshold to serve as the primary balanced classifier.

6.2.4 Implementation Guidance

- Deploy Random Forest with SMOTE as the primary classifier when the objective is a balanced shortlist of likely subscribers.
- Use Original XGBoost probabilities to rank customers when marketing capacity is defined as a fixed number or percentage of customers.
- Do not assume that 0.50 is the final operational threshold. Test thresholds against calling capacity, expected conversion value and the cost of unnecessary calls.
- Continue excluding duration for pre-call use. Duration may be analysed after calls for quality monitoring, but it should not enter the targeting score.
- Monitor performance by month, contact channel and customer segment because campaign conditions may change over time.
- Retrain and revalidate the model when new campaign data become available; future work should add temporal validation and probability calibration.

6.2.5 Final Selection

Random Forest with SMOTE should be presented as the best balanced classification model, while Original XGBoost should be retained as the strongest probability-ranking alternative. This two-model

recommendation is more defensible than claiming that one model is universally best for every campaign objective.

7.0 Conclusion

This study utilized descriptive analytics, anomaly analysis and predictive modelling on the UCI Bank Marketing dataset with the aim of studying the customers behaviour and predicting customer subscription to term deposit accounts. There were 17 variables and 45,211 customer records. A large class imbalance was seen, with only 5289 customers subscribed, and a positive subscription rate of 11.70%.

Descriptive analysis revealed significant differences between the customer groups. There were relatively high rates of subscribing among the students and retirees compared to the overall campaign average. The analysis also showed that previous campaign outcome, method of contact, housing-loan status, campaign month, age and balance, as well as the previous campaign activity were predictive signals.Five major anomalies were examined. These consisted of extreme account balance, excessive campaign contact, high previous events unknown ratio, the use of -1 in pdays and extremely long call durations. The results indicated that anomalies should not be eliminated without any consideration as some irrational values were legitimate customer attributes, or significant business situations.

Four machine-learning algorithms were tested with both the original and SMOTE-balanced training data sets. When it comes to the highest precision in terms of subscribers, Original Logistic Regression enjoys the crown, while Logistic Regression with SMOTE has the highest recall in terms of subscribers. Random Forest using SMOTE had the highest conversion F1 score of 44.98%, and it offered the best balance between conversion quality and subscribers covered. Original XGBoost had the highest ROC-AUC (80.31%) and PR-AUC (46.87%), which is the best probability-ranking model. According to the results, it is suggested that the Random Forest with SMOTE can be used as the main balanced classification model. If the capacity of the campaign is limited, and customers need to be prioritised based on the likelihood of subscribing then an alternative ranking model should be kept, which is original XGBoost.

The results of the study show how banks can use machine learning to minimize wasted customer contact, optimize resources, and boost campaign success. Predictive models should be used in conjunction with managerial judgement, however. Pre-implementation considerations for model interpretation, customer privacy, fairness and operational practicality. Probability calibration, threshold optimisation, other class-balancing approaches, as well as explainable artificial-intelligence methods could be explored in future studies. A more comprehensive evaluation of the models should also be conducted on more recent campaigns and to assess the stability of the models over time. The creation of a live customer-scoring platform could also benefit real-time campaign management and ongoing monitoring of models.

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